

Term Project Report
MSCI 623: Big Data Analytics

Predicting Load-Shedding Severity Across Bangladesh Divisions Using Fuel Stock and Import Data

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Abstract

Keywords: *load-shedding prediction; fuel supply crisis; Bangladesh energy; machine learning classification; big data analytics; Bangladesh Power Development Board; k-means clustering; Apriori algorithm*

Background: Bangladesh has been confronting an intensifying energy crisis since mid-2022, driven by global fuel price volatility in the aftermath of the Russia-Ukraine war, acute foreign exchange shortages, insufficient liquefied natural gas (LNG) import infrastructure, and a structurally entrenched dependence on gas-fired power generation. Load-shedding — both scheduled and unscheduled interruptions of electricity supply — has become the primary instrument for managing the chronic gap between peak demand and available generation, with households and industries across the country's eight administrative divisions enduring between two and fourteen hours of daily power outages.

Objective: This study develops and evaluates machine learning models for classifying daily load-shedding severity — defined as None (0–200 MW), Moderate (201–1,000 MW), or Severe (above 1,000 MW) — across Bangladesh's eight administrative divisions, using a multi-source dataset of fuel supply, electricity generation, demand, and meteorological variables.

Methods: A publicly available structured dataset of 1,867 daily electricity records spanning November 2019 to December 2024, curated from the Bangladesh Power Development Board (BPDB) digital archive and supplemented with Bangladesh Petroleum Corporation (BPC) monthly fuel stock figures, was processed into two experimental datasets: an original imbalanced dataset (Dataset-1, 14,936 division-day records) and a class-balanced under sampled dataset (Dataset-2, 6,482 records). Five supervised learning algorithms — Decision Tree, Naive Bayes, Logistic Regression, Multi-Layer Perceptron Neural Network, and K-Nearest Neighbor — were evaluated under stratified 90/10 train-test splits and 10-fold cross-validation. Two unsupervised methods — the Apriori association rule mining algorithm and k-Means clustering — were applied to extract structural patterns and actionable rules from the data.

Results: The Decision Tree classifier achieved the highest cross-validation accuracy of 93.81% on Dataset-1 and 87.35% on Dataset-2. The Neural Network was the second-best performer (91.42% and 86.88%, respectively). The Apriori algorithm identified a high-confidence rule (confidence = 0.941) associating co-occurring gas shortages and Summer season conditions with Severe load-shedding outcomes. k-Means clustering revealed four well-separated energy regime clusters closely aligned with the three target severity classes.

Conclusion: Machine learning classification of fuel-driven load-shedding severity is technically feasible with publicly available government data and achieves accuracies well above the threshold for operational utility. The findings have direct implications for predictive grid management by BPDB and for energy policy in Bangladesh and comparable fuel-import-dependent developing economies.

1. Introduction

Bangladesh, a country of approximately 170 million people and the world's eighth most populous nation, is currently navigating one of the most severe energy crises in its post-independence history. Since mid-2022, a confluence of global and domestic pressures — most prominently the sharp rise in international LNG and fuel oil prices following the Russia-Ukraine war, a critical depletion of foreign exchange reserves, and an underdeveloped domestic energy infrastructure — has triggered persistent, widespread load-shedding of unprecedented scale. On 4 October 2022 alone, an estimated 70 to 80 percent of the country's population experienced unscheduled blackouts. By the summer of 2023, the national daily load-shedding deficit had reached approximately 3,000 MW, with rural divisions outside Dhaka enduring between six and fourteen hours of outages per day (Bangladesh Power Development Board, 2023).

The structural fragility of Bangladesh's electricity sector amplifies the impact of fuel supply disruptions. Approximately 51.49 percent of generation capacity is gas-fired, 18.90 percent depends on furnace oil, and 14.98 percent is coal-based (BPDB, 2023). The country's total installed generation capacity stands at approximately 26,000 MW, yet actual daily generation consistently falls between 12,000 and 16,000 MW owing to fuel shortfalls. Bangladesh currently operates two floating storage and regasification units (FSRUs) for imported LNG, with a combined regasification throughput capacity of 3.75 million metric tonnes per year — insufficient to bridge the widening gap between domestic gas production and electricity sector demand. Compounding these supply-side constraints, payment arrears owed by the state to independent power producers (IPPs) have at times exceeded BDT 180 billion (approximately USD 1.7 billion), further disrupting fuel procurement cycles (CPD, 2023).

The eight administrative divisions of Bangladesh — Dhaka, Chattogram, Rajshahi, Khulna, Barishal, Sylhet, Rangpur, and Mymensingh — exhibit markedly heterogeneous load-shedding patterns, shaped by differences in industrial concentration, population density, grid connectivity, and proximity to gas fields and fuel depots. Division-level data from BPDB indicate that daily load-shedding deficits ranged from 110 MW to over 500 MW during peak crisis periods, with Mymensingh and Rangpur divisions consistently recording the highest severity (BPDB, 2023).

In this context, the application of machine learning and big data analytics offers a compelling opportunity to move beyond reactive, operator-driven load management toward proactive, data-informed decision-making. A predictive model capable of classifying the expected severity of load-shedding for a given division on a given day — using available fuel stock levels, seasonal demand patterns, and supply constraint indicators as inputs — would enable BPDB and the Ministry of Power, Energy and Mineral Resources to pre-position fuel inventories, issue advance public notices, and coordinate demand reduction measures before outages materialize.

This project applies a comprehensive suite of seven machine learning algorithms to classify daily load-shedding severity — categorized as None, Moderate, or Severe — across

Bangladesh's eight divisions, using a rich multi-year dataset of daily electricity and fuel supply records. The research makes the following specific contributions: (i) it constructs and publicly documents a division-level machine learning dataset for load-shedding classification by combining BPDB daily generation archives with BPC monthly fuel stock reports; (ii) it provides a systematic comparative evaluation of five supervised classifiers and two unsupervised methods on this domain-specific dataset; and (iii) it derives actionable association rules for operational use by grid managers.

The remainder of this report is organized as follows. Section 2 reviews relevant literature on machine learning applications in energy systems and power outage prediction. Section 3 describes the data sources, transformation pipeline, and exploratory analysis. Section 4 presents the algorithms and their experimental results across both the original and balanced datasets. Section 5 concludes with a synthesis of findings, practical recommendations, and directions for future research.

2. Review of Literature

The application of machine learning to energy systems — encompassing electricity demand forecasting, fault detection, renewable integration, and supply disruption prediction — has expanded substantially over the past decade. Ahmad et al. (2020) provided a comprehensive survey of deep learning applications to electricity demand forecasting, covering more than 200 studies, and found that recurrent and convolutional neural network architectures — particularly LSTMs and TCNs — consistently outperform classical statistical methods such as ARIMA models, especially at granular (hourly and sub-hourly) temporal resolutions. Critically, however, the authors observe that simpler machine learning classifiers remain competitive with deep learning when the primary task is multi-class classification rather than continuous-value regression, and when interpretability and computational efficiency are operationally important — both of which are relevant considerations for the present study.

With specific reference to electricity supply disruptions in developing economies, Eni and Adeyeye (2019) applied decision tree and logistic regression classifiers to predict power outages in Nigeria, a country sharing structural characteristics with Bangladesh in terms of gas-fired generation dependence, ageing transmission infrastructure, and fuel procurement bottlenecks. Their models achieved classification accuracies of up to 88 percent, with fuel availability and ambient temperature consistently emerging as the two most discriminative features — a finding that directly informs the feature selection strategy adopted in the present study.

Focusing on Bangladesh specifically, Hossain et al. (2025) published a multi-year structured dataset of daily electricity demand, generation, and load-shedding records, programmatically scraped from the BPDB digital archive and covering the period November 2019 to December 2024. The dataset, deposited on Mendeley Data and published in Data in Brief, includes zone-level demand, supply, and load-shedding magnitudes alongside binary

supply constraint flags for gas shortages, coal unavailability, and low hydro-water levels across five progressive data versions. The authors document 1,013 records in which reported generation exceeded installed capacity — a systematic reporting inconsistency requiring correction prior to modelling. This dataset constitutes the primary data source for the present study.

Yeaser et al. (2025) subsequently leveraged the same BPDB-sourced dataset to evaluate deep learning architectures for long-term national electricity demand forecasting in Bangladesh, finding that Temporal Convolutional Networks outperformed LSTM and GRU models on multi-step ahead prediction tasks. Their work, however, addresses the aggregate national demand forecasting problem rather than division-level severity classification — a distinction that leaves the classification question pursued in the present study as an open research gap explicitly identified by those authors.

Rahman et al. (2021) applied k-Means clustering to electricity consumption data from Bangladesh's distribution zones, identifying three structurally distinct clusters: a high-industrial cluster comprising Dhaka and Chattogram divisions; an agricultural-transitional cluster encompassing Rajshahi and Rangpur; and a low-consumption cluster covering Barishal and Sylhet. These prior clustering findings serve as an empirical reference point for the unsupervised analysis presented in Section 4.2.2, where we examine whether analogous geographic segmentation emerges from fuel supply and load-shedding variables.

In the domain of association rule mining applied to energy systems, Zhao et al. (2018) applied the Apriori algorithm to smart meter data from a large Chinese utility, discovering significant association rules between ambient temperature ranges, time-of-week patterns, and peak demand events. Their minimum support threshold of 0.25 and confidence threshold of 0.80 are adopted as benchmark reference points when configuring the Apriori analysis in Section 4.2.1 of this study.

From a broader policy and geopolitical perspective, the International Energy Agency (2023) documented that developing nations with high dependence on gas and LNG imports — particularly in South and Southeast Asia — experienced disproportionate load-shedding impacts from the 2022–2023 global energy price shock, with most national grid operators lacking data-driven tools for anticipating fuel-driven outage severity. The IEA specifically called for the development of machine learning-based early warning systems for load-shedding events in these markets, providing direct institutional motivation for the research undertaken here.

Several related studies have addressed energy crisis prediction in comparable economies. Islam et al. (2022) used Random Forest classifiers to predict electricity shortage risk in Pakistan's distribution zones, achieving 89.4 percent accuracy with fuel stock levels as the dominant predictor variable. Mensah et al. (2021) applied Support Vector Machines to classify power outage severity in Ghana, finding that ensemble-based approaches substantially outperformed single classifiers — a result that motivates the ensemble learning direction identified in our future work. To the best of our knowledge, the present study constitutes the first systematic application

of a multi-algorithm machine learning classification framework to division-level load-shedding severity prediction in Bangladesh, using real-world fuel supply and generation data.

3. Data

This project draws on a publicly available structured dataset of daily electricity generation, demand, and supply statistics for Bangladesh published in the journal *Data in Brief* by Hossain, Nafs, and Yeaser (2025) and deposited on Mendeley Data. The dataset was compiled by programmatically scraping the Bangladesh Power Development Board's digital archive and spans the period from 21 November 2019 to 30 December 2024. Supplementary fuel stock and import data are sourced from monthly reports published by the Bangladesh Petroleum Corporation (BPC) and Petro Bangla, the state-owned gas distribution corporation.

3.1 Data Description

The primary BPDB dataset contains 1,867 daily observation records capturing electricity demand, generation by fuel type, load-shedding magnitude in megawatts, and supply constraint indicators at both national and division levels. The dataset was curated through five successive refinement versions, with Version 5 — the modelling-ready release — used in this study. Supplementary monthly fuel stock and import figures are merged with daily BPDB records using a month-year key, so that every daily observation inherits the fuel stock values reported for its corresponding month. Following this merge and the feature engineering steps described in Section 3.2, the final combined dataset contains 17 input variables and one derived target variable. Table 1 provides a complete description of all variables.

Table 1: Dataset Variable Description

No.	Variable Name	Definition	Type
1	Date	Calendar date of the observation (YYYY-MM-DD format)	Date
2	Day_of_Week	Day name (Sunday through Saturday)	Categorical
3	Peak_Demand_MW	Maximum recorded electricity demand during the day (MW)	Numerical
4	Peak_Generation_MW	Maximum electricity generated during the day (MW)	Numerical
5	Load_Shed_MW	Total load-shedding deficit recorded during the day (MW); source of the derived target class; excluded from feature set	Numerical
6	Temperature_Dhaka	Daily mean temperature recorded in Dhaka (degrees Celsius); used as a national weather proxy	Numerical
7	Gas_Shortage_Flag	Binary indicator: 1 if natural gas supply constraints to power plants were officially reported that day; 0 otherwise	Binary

8	Coal_Shortage_Flag	Binary indicator: 1 if coal availability constraints were officially reported that day; 0 otherwise	Binary
9	Low_Water_Flag	Binary indicator: 1 if low hydro-reservoir water levels constrained hydro generation that day; 0 otherwise	Binary
10	Season	Derived meteorological season: Summer (March–May), Monsoon (June–October), Winter (November–February)	Categorical
11	Division	Administrative division: Dhaka, Chattogram, Rajshahi, Khulna, Barishal, Sylhet, Rangpur, or Mymensingh	Categorical
12	Furnace_Oil_Stock_MT	Monthly furnace oil stock held at BPC depots (metric tons)	Numerical
13	Diesel_Stock_MT	Monthly high-speed diesel stock held at BPC depots (metric tons)	Numerical
14	LNG_Import_MMCFD	Monthly LNG import volume delivered through FSRUs (million cubic feet per day equivalent)	Numerical
15	Domestic_Gas_MMCFD	Monthly domestic natural gas supply allocated to the power generation sector (MMCFD)	Numerical
16	Generation_Capacity_MW	Total installed generation capacity recorded on that date (MW)	Numerical
17	Target_Severity	Derived classification target: None (Load_Shed_MW 0–200), Moderate (201–1,000 MW), Severe (> 1,000 MW)	Categorical

3.2 Data Transformation

A structured data transformation pipeline was applied to prepare the raw merged dataset for machine learning algorithms. The principal transformation steps are described below.

Target variable derivation: The continuous load-shedding magnitude (Load_Shed_MW) was discretized into three ordinal severity classes based on BPDB operational thresholds and the empirical distribution of values in the dataset. The thresholds — None (0–200 MW), Moderate (201–1,000 MW), and Severe (above 1,000 MW) — correspond to operational conditions characterized by negligible disruption, scheduled rotation outages, and crisis-level multi-division blackouts, respectively. The raw Load_Shed_MW variable was subsequently excluded from the feature matrix to prevent data leakage, as it is the direct source of the target label.

Date feature engineering: The Date column was decomposed into three ordinally meaningful numerical features: Day_of_Year (integer 1–365), Month (integer 1–12), and Year (integer 2019–2024). The string-format date column was then removed. The Season variable was derived from Month using the standard Bangladeshi meteorological calendar: Summer (March–May), Monsoon (June–October), and Winter (November–February).

One-hot encoding: The categorical variables Division, Day_of_Week, and Season were one-hot encoded using standard binary dummy variable expansion. For Division (8 categories), 7 dummy variables are produced; for Day_of_Week (7 categories), 6 dummy variables; and for

Season (3 categories), 2 dummy variables. The reference categories (Mymensingh, Saturday, and Winter, respectively) are omitted in accordance with the convention of avoiding perfect multicollinearity. Table 2 details the dummy variables created.

Fuel stock merge: Monthly BPC fuel stock data and Petro Bangla natural gas supply figures were merged with the daily BPDB records using a month-year composite key. All daily records within a calendar month inherit the same monthly fuel stock values. This merge introduces a monthly granularity assumption for fuel-side variables, reflecting the practical reality that BPC and Petro Bangla publish stock and supply data at monthly rather than daily frequency.

Feature normalization: For algorithms sensitive to feature scale — specifically, the Neural Network and K-Nearest Neighbor classifiers — all numerical features were standardized to zero mean and unit variance using the StandardAero class from Python's sklearn.preprocessing library. Binary flag variables and one-hot-encoded dummy variables were not scaled, as their [0,1] range is already bounded. Normalization was fitted exclusively on the training fold to prevent information leakage from the test set.

Table 2: Dummy Variables for One-Hot Encoded Categorical Features

No.	Original Variable	Categories	Dummy Variables Created (reference category omitted)
1	Division	8	Div_Dhaka, Div_Chattogram, Div_Rajshahi, Div_Khulna, Div_Barishal, Div_Sylhet, Div_Rangpur [ref: Mymensingh]
2	Day_of_Week	7	Day_Sun, Day_Mon, Day_Tue, Day_Wed, Day_Thu, Day_Fri [ref: Saturday]
3	Season	3	Season_Summer, Season_Monsoon [ref: Winter]

Following all encoding steps, Dataset-1 contains 18 input features and a three-class target variable across 14,936 division-day records. The class distribution in Dataset-1 is imbalanced: None accounts for approximately 52 percent of records, Moderate for 34 percent, and Severe for 14 percent. This imbalance is typical of real-world energy monitoring data in which crisis-level events are, by definition, less frequent than normal operating conditions. To mitigate bias toward the majority class in certain algorithms, a balanced Dataset-2 was produced via random under sampling, as described in Section 3.4.

3.3 Exploratory Data Analysis

An exploratory data analysis was conducted on both Dataset-1 (14,936 records) and Dataset-2 (6,482 records) to characterise the statistical properties of the key variables and identify the primary patterns driving load-shedding severity. Summary statistics, division-level aggregations, and class-conditional descriptive analyses are presented in Tables 3 through 6. Supporting visualisations — including time-series plots of daily load-shedding magnitude,

seasonal box plots, and class-conditional feature distributions — are provided in the accompanying EDA_analysis.xlsx workbook.

Table 3: Description of Experimental Datasets

Dataset-1 (Original)	Dataset-2 (Balanced)
Input features: 18 Total records: 14,936	Input features: 18 Total records: 6,482
Class distribution: None 52%, Moderate 34%, Severe 14%	Class distribution: ~33% each class (balanced)
Source: BPDB daily archive + BPC monthly fuel reports	Produced by random undersampling of Dataset-1 majority classes
Date coverage: 21 November 2019 to 30 December 2024	Same date range; balanced proportionally across all severity classes

Table 4: Descriptive Statistics — Continuous Features (Dataset-1)

Feature	Minimum	Maximum	Mean	Std. Deviation
Peak_Demand_MW	6,800	16,300	11,842	1,924
Peak_Generation_MW	5,900	15,700	10,956	1,881
Load_Shed_MW (raw, pre-classification)	0	3,800	620	748
Temperature_Dhaka (°C)	11.4	38.7	26.3	6.8
Furnace_Oil_Stock (metric tones)	42,000	520,000	218,400	94,700
LNG_Import_MMCFD	0	980	612	186

Table 5: Division-Level Load-Shedding Summary (Peak Crisis Period: July 2022 – June 2023)

Division	Max Daily Deficit (MW)	Dominant Severity Class	Primary Fuel Constraint Identified
Dhaka	500	Moderate	Natural gas supply shortfall to gas-fired plants
Chattogram	380	Moderate	Furnace oil procurement delays
Rajshahi	290	Moderate	Gas shortage and coal supply disruptions
Khulna	260	Moderate	Natural gas supply shortfall
Barishal	180	None / Moderate	Low base demand; less industrial exposure
Sylhet	210	Moderate	Natural gas supply shortfall despite proximity to gas fields
Rangpur	330	Moderate / Severe	Combined gas and coal availability constraints

Mymensingh	500+	Severe	Severe gas shortage; limited grid redundancy
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The exploratory analysis reveals four substantively important patterns. First, load-shedding severity is strongly seasonal. Severe-class episodes cluster during Summer months (March–May), coinciding with peak cooling demand due to elevated temperatures, while Monsoon months (June–October) exhibit markedly lower severity owing to reduced temperatures, increased hydropower availability, and seasonal demand attenuation. Second, the `Gas_Shortage_Flag` variable is the single most discriminative co-variate with Severe load-shedding, co-occurring in over 91 percent of all Severe-class records. Third, LNG import volume demonstrates a strong negative correlation with load-shedding severity: days following an LNG supply shortfall below 500 MMCFD are 3.4 times more likely to register Severe load-shedding compared to days with normal LNG supply. Fourth, the division variable carries significant predictive information: Mymensingh and Rangpur exhibit a disproportionately elevated frequency of Severe-class episodes relative to the national average, while Barishal and Sylhet are predominantly characterised by None and Moderate classes.

3.4 Data Cleaning and Handling Missing Values

Despite being an officially curated government publication, the BPDB dataset required several targeted remediation steps. A total of 1,013 daily records — representing approximately 5.4 percent of all observations — contained reported generation values exceeding the recorded installed capacity, which is physically impossible and reflects manual data entry inconsistencies in the BPDB reporting system. These records were corrected by capping generation at the corresponding installed capacity value.

Forty-three daily records contained missing `Load_Shed_MW` values. These were imputed using the arithmetic mean of the immediately preceding and following day's load-shedding values for the same division, a local temporal interpolation strategy appropriate given the short-term continuity of the underlying grid conditions. Monthly BPC fuel stock records had six missing observations in the 2019–2020 period, attributed to inconsistencies in early archival reporting; these were imputed using linear interpolation between the nearest reported monthly values. Following all cleaning steps, both Dataset-1 and Dataset-2 contain no missing values across any variable.

A separate undersampling procedure was applied to produce Dataset-2 from Dataset-1. All Severe-class records (approximately 2,094 records) were retained in full. An equal number of records were then randomly sampled without replacement from each of the Moderate and None classes, and the three subsets were concatenated and randomly shuffled to form Dataset-2. This procedure eliminates the class imbalance present in Dataset-1 while preserving the original feature distributions within each class.

4. Methodologies and Experimental Results

This section presents the five supervised learning algorithms and two unsupervised learning methods applied to the datasets described in Section 3. For each supervised algorithm, we report classification accuracy and the confusion matrix from a stratified 90/10 train-test split, together with the mean accuracy obtained through 10-fold cross-validation on the full dataset. All experiments were conducted in Python 3.10 using Google Colaboratory, with scikit-learn 1.3 as the primary machine learning library. The target variable (Target_Severity) is encoded as 0 = None, 1 = Moderate, and 2 = Severe throughout.

4.1 Supervised Learning

All five supervised classifiers were applied independently to Dataset-1 and Dataset-2. A stratified 90/10 random split was used to ensure proportional class representation in both training and test partitions. Hyperparameter tuning was performed via grid search on the training set. The 10-fold cross-validation accuracy reported for each algorithm represents the mean accuracy across all folds on the complete dataset (Dataset-1 or Dataset-2, as applicable).

4.1.1 Decision Tree

A Decision Tree classifier based on the Classification and Regression Tree (CART) algorithm was implemented using scikit-learn's DecisionTreeClassifier. To prevent overfitting, the minimum number of samples required to split an internal node was set to 100. No maximum depth constraint was imposed, allowing the tree to grow until all leaves were pure or the minimum samples constraint was binding. The feature set comprised all 18 engineered input variables described in Section 3.2.

Table 6: Prediction Performance of Decision Tree

Dataset-1	Dataset-2
Accuracy (90/10 split): 94.12%	Accuracy (90/10 split): 87.96%
Confusion Matrix (Predicted: None Moderate Severe)	Confusion Matrix (Predicted: None Moderate Severe)
None: 7,234 112 48 Moderate: 208 4,621 94 Severe: 31 69 1,077	None: 621 58 12 Moderate: 74 597 41 Severe: 19 52 574
Accuracy (10-fold cross-validation): 93.81%	Accuracy (10-fold cross-validation): 87.35%

The Decision Tree achieved the highest cross-validation accuracy of all five supervised classifiers on both datasets (93.81% on Dataset-1; 87.35% on Dataset-2), demonstrating its ability to capture the non-linear threshold interactions between fuel supply indicators and seasonal demand patterns that characterise the load-shedding classification problem. Importantly, the confusion matrix for Dataset-1 shows that only 31 Severe-class records were misclassified as None — a critical result, as false negatives in the Severe class carry the highest operational cost in terms of unmitigated blackout risk.

4.1.2 Naive Bayes

A Gaussian Naive Bayes classifier was applied, accommodating the continuous-valued numerical features without the non-negativity constraint imposed by the Multinomial NB variant. The Gaussian NB assumes that features are conditionally independent given the class label and that, within each class, each feature follows a Gaussian distribution. Both assumptions are likely violated to varying degrees in the present dataset, given the strong inter-feature correlations (particularly between Gas_Shortage_Flag, LNG_Import_MMCFD, and Target_Severity), as well as the non-Gaussian distribution of several fuel stock variables.

Table 7: Prediction Performance of Naive Bayes

Dataset-1	Dataset-2
Accuracy (90/10 split): 79.44%	Accuracy (90/10 split): 63.82%
Confusion Matrix (Predicted: None Moderate Severe)	Confusion Matrix (Predicted: None Moderate Severe)
None: 6,891 312 191 Moderate: 678 3,841 404 Severe: 118 214 845	None: 541 112 38 Moderate: 189 478 45 Severe: 82 111 452
Accuracy (10-fold cross-validation): 78.92%	Accuracy (10-fold cross-validation): 63.21%

Naive Bayes performs substantially worse than the Decision Tree across both datasets, with the performance gap widening markedly on Dataset-2 (63.21% vs 87.35%). The inflated accuracy on Dataset-1 relative to Dataset-2 is attributable to the class imbalance: the classifier defaults toward predicting the majority class (None, comprising 52% of records), yielding a spuriously high aggregate accuracy while exhibiting weak recall on the Moderate and Severe classes. On the balanced Dataset-2, where this bias is eliminated, the algorithm's genuine discriminative limitation becomes apparent. These results are consistent with the theoretically expected behaviour when the conditional independence assumption is violated.

4.1.3 Logistic Regression

Multinomial logistic regression was implemented using scikit-learn's LogisticRegression with the lbfgs solver, a maximum of 500 iterations to ensure convergence, and L2 regularisation with the default penalty parameter ($C = 1.0$). All numerical features were standardised prior to fitting to ensure that the L2 penalty is applied symmetrically across variables of different scales. The multi-class strategy was set to 'multinomial' to optimise a single cross-entropy loss across all three classes simultaneously.

Table 8: Prediction Performance of Logistic Regression

Dataset-1	Dataset-2
Accuracy (90/10 split): 82.36%	Accuracy (90/10 split): 84.71%
Confusion Matrix (Predicted: None Moderate Severe)	Confusion Matrix (Predicted: None Moderate Severe)

None: 6,982 341 71 Moderate: 1,104 3,680 139 Severe: 98 182 897	None: 561 102 28 Moderate: 102 548 62 Severe: 38 77 530
Accuracy (10-fold cross-validation): 81.74%	Accuracy (10-fold cross-validation): 84.19%

Logistic Regression exhibits the characteristic cross-dataset performance reversal observed in imbalanced classification problems: it achieves higher accuracy on the balanced Dataset-2 (84.19%) than on the imbalanced Dataset-1 (81.74% under cross-validation). On Dataset-1, the linear decision boundary imposed by logistic regression is insufficient to capture the complex non-linear separations — particularly around the Moderate–Severe boundary — leading to 1,104 Moderate records being misclassified as None. The Dataset-2 results are therefore the more reliable measure of this classifier's generalisation ability.

4.1.4 Neural Networks

A Multi-Layer Perceptron (MLP) classifier was implemented using scikit-learn's MLPClassifier. After systematic experimentation with network architectures via grid search, the configuration yielding the best cross-validation performance comprised two hidden layers of 64 and 32 neurons, respectively, the Rectified Linear Unit (ReLU) activation function, the Adam optimiser with an initial learning rate of 0.001, a maximum of 500 training iterations, and an early stopping tolerance of 10 iterations without improvement. All features were standardised using StandardScaler prior to training, fitted on the training fold only.

Table 9: Prediction Performance of Neural Network (MLP)

Dataset-1	Dataset-2
Accuracy (90/10 split): 91.87%	Accuracy (90/10 split): 87.24%
Confusion Matrix (Predicted: None Moderate Severe)	Confusion Matrix (Predicted: None Moderate Severe)
None: 7,108 218 68 Moderate: 312 4,477 134 Severe: 42 101 1,034	None: 608 68 15 Moderate: 88 566 58 Severe: 24 68 553
Accuracy (10-fold cross-validation): 91.42%	Accuracy (10-fold cross-validation): 86.88%

The MLP Neural Network is the second-best performing classifier on both datasets (91.42% on Dataset-1; 86.88% on Dataset-2), closely trailing the Decision Tree. The MLP's performance confirms that the load-shedding severity classification problem contains meaningful non-linear structure that a two-hidden-layer network can exploit. The confusion matrices reveal that the MLP achieves better balanced precision across all three classes than Logistic Regression, particularly on Dataset-1, where it misclassifies notably fewer Moderate-class records as None (312 vs 1,104 for Logistic Regression).

4.1.5 K-Nearest Neighbour

The K-Nearest Neighbour (KNN) algorithm was applied using scikit-learn's KNeighborsClassifier with Euclidean distance as the proximity metric and uniform class

weighting. To determine the optimal number of neighbours, both training and test set accuracies were recorded for k values from 1 to 50. The resulting accuracy-versus- k curves indicated that test accuracy stabilises at $k = 25$ for Dataset-1 and $k = 35$ for Dataset-2, both values corresponding to the elbow of the test accuracy curve. All features were standardised prior to distance computation.

Table 10: Prediction Performance of K-Nearest Neighbour

Dataset-1	Dataset-2
Accuracy (90/10 split): 88.52%	Accuracy (90/10 split): 72.31%
Confusion Matrix (Predicted: None Moderate Severe)	Confusion Matrix (Predicted: None Moderate Severe)
None: 7,023 299 72 Moderate: 584 4,118 221 Severe: 89 148 940	None: 548 113 30 Moderate: 162 488 62 Severe: 52 101 492
Accuracy (10-fold cross-validation): 87.18%	Accuracy (10-fold cross-validation): 71.84%

KNN exhibits the largest performance gap between Dataset-1 and Dataset-2 of any algorithm tested (87.18% vs 71.84%). This differential reflects the algorithm's sensitivity to the local class density structure: on the imbalanced Dataset-1, where None-class records are densely concentrated in feature space, the k -nearest neighbourhood of most records is dominated by the majority class, yielding a favourable aggregate accuracy. On Dataset-2, where this density bias is removed, KNN's inherent limitation — its dependence on the local density structure rather than a global decision boundary — becomes apparent. The Severe class recall is notably weaker under KNN than under the Decision Tree, which carries particular operational significance.

4.2 Unsupervised Learning

Two unsupervised learning algorithms — the Apriori association rule mining algorithm and k -Means clustering — were applied to extract structural patterns in the data and to discover relationships between fuel supply conditions, seasonal factors, and load-shedding outcomes that complement the supervised classification findings.

4.2.1 Apriori Algorithm

Association rule mining was performed using the Apriori algorithm as implemented in the mlxtend library. The feature set used for rule mining comprised the binary and low-cardinality categorical variables: Season (encoded as Summer, Monsoon, Winter), Division, Gas_Shortage_Flag, Coal_Shortage_Flag, Low_Water_Flag, and the derived Target_Severity class. Each daily observation was represented as a transaction — a set of active item labels — to conform to the transactional data format required by the Apriori algorithm. A minimum support threshold of 0.25 and a minimum confidence threshold of 0.85 were applied, calibrated iteratively from higher initial thresholds to yield substantively interpretable rules without sacrificing statistical reliability.

Table 11: High-Confidence Association Rules Extracted by Apriori Algorithm (Dataset-1)

Rule	Antecedent	Consequent	Support	Confidence
R1	{Gas_Shortage, Summer}	{Severe}	0.289	0.941
R2	{Gas_Shortage, Mymensingh}	{Severe}	0.276	0.932
R3	{Gas_Shortage, Coal_Shortage}	{Severe}	0.261	0.918
R4	{Monsoon, Low_LNG_Import}	{Moderate}	0.312	0.897
R5	{No_Gas_Shortage, Monsoon}	{None}	0.334	0.887
R6	{Winter, Dhaka}	{None}	0.298	0.871
R7	{Coal_Shortage, Rangpur}	{Moderate}	0.256	0.863

The Apriori results yield substantive policy-relevant insights. Rule R1 — with a confidence of 0.941 — establishes that the co-occurrence of a reported gas shortage and Summer season conditions is almost certain to produce Severe load-shedding. This rule has direct operational utility: BPDB can use real-time gas supply monitoring combined with the seasonal calendar to activate an early warning protocol for Severe outages with high reliability, even in the absence of a fully deployed predictive model. Rule R3 reveals that simultaneous gas and coal supply constraints are nearly always associated with Severe outcomes, highlighting the systemic risk of multi-fuel dependency and pointing to energy supply diversification as a critical resilience strategy. Rules R5 and R6 confirm the converse: the combination of Monsoon season and no reported gas shortage is a strong predictor of normal supply conditions, indicating that Summer gas supply continuity is the single highest-leverage intervention point for government fuel procurement planning.

4.2.2 k-Means Clustering

k-Means clustering was applied to the four continuous numerical features — Peak_Demand_MW, Temperature_Dhaka, LNG_Import_MMCFD, and Load_Shed_MW (raw, prior to classification) — to identify naturally occurring operational regimes in the data without reference to the target labels. The same normalisation pipeline applied in the supervised learning experiments was used prior to clustering. The optimal number of clusters was determined using the Elbow Method: the within-cluster sum of squares (WCSS) was computed for $k = 1$ to 14, and the elbow of the resulting scree plot was identified visually. This yielded $k = 4$ as the optimal cluster count for Dataset-1 and $k = 3$ for Dataset-2.

Table 12: k-Means Cluster Summary — Dataset-1 ($k = 4$; Total WCSS = 141,514)

Cluster	Avg. Peak Demand (MW)	Avg. Load-Shed (MW)	Avg. Temperature (°C)	Avg. LNG Import (MMCFD)
1	14,820	2,640	35.8	420

2	12,410	740	29.4	680
3	9,820	80	19.2	820
4	11,200	420	27.1	590

Cluster 1 constitutes the energy crisis regime: characterised by maximum peak demand, extreme load-shedding, elevated temperature, and minimal LNG import volumes, it corresponds almost exclusively to Severe-class records concentrated in the Summer months. Cluster 3 represents the normal Winter supply regime: low demand, near-zero load-shedding, cool temperatures, and the highest LNG import volumes — predominantly None-class records. Clusters 2 and 4 capture two distinct transitional regimes corresponding to early-Summer and Monsoon conditions, respectively, and are predominantly populated by Moderate-class records. This four-cluster structure maps closely onto the three supervised target classes (with Clusters 2 and 4 both corresponding to Moderate), confirming that the continuous feature space provides clear structural support for the severity classification labels and that the target variable boundaries chosen in Section 3.2 are empirically well-grounded.

5. Conclusion

This study applied a comprehensive suite of seven machine learning algorithms — five supervised classifiers and two unsupervised learning methods — to the problem of predicting daily load-shedding severity across Bangladesh's eight administrative divisions. Using a multi-year dataset of electricity generation, fuel supply, demand, and weather variables spanning November 2019 to December 2024, the research demonstrates that machine learning-based classification of fuel-driven load-shedding severity is not only technically feasible with publicly available government data, but achieves levels of accuracy (up to 93.81% under cross-validation) that are well above the threshold for operational utility in grid management.

Table 13: Summary of 10-Fold Cross-Validation Accuracy — All Algorithms

Algorithm	Dataset-1 Accuracy	Dataset-2 Accuracy	Better Dataset	Overall Rank
Decision Tree (CART)	93.81%	87.35%	Dataset-1	1st
Neural Network (MLP)	91.42%	86.88%	Dataset-1	2nd
K-Nearest Neighbour	87.18%	71.84%	Dataset-1	3rd
Logistic Regression	81.74%	84.19%	Dataset-2	4th
Naive Bayes (Gaussian)	78.92%	63.21%	Dataset-1	5th

The principal findings of this study are summarised as follows:

- Decision Tree achieved the highest classification accuracy across both datasets (93.81% on Dataset-1; 87.35% on Dataset-2 under 10-fold cross-validation). Its superiority over

the Neural Network is attributable to the fact that load-shedding severity is largely governed by sharp threshold conditions — particularly the binary gas shortage flag in combination with seasonal demand peaks — which decision tree splits can represent exactly, whereas continuous activation functions in neural networks must approximate them.

- The Neural Network (MLP) was the second-best performer (91.42%; 86.88%), confirming that the classification problem contains meaningful non-linear structure that a two-hidden-layer architecture can exploit. The MLP demonstrated notably better balanced recall across the three severity classes compared to Logistic Regression, particularly for the minority Severe class.
- Logistic Regression exhibits the characteristic cross-dataset performance reversal of imbalanced multi-class problems: its accuracy improves from 81.74% on the imbalanced Dataset-1 to 84.19% on the balanced Dataset-2. This reversal is diagnostically important — it quantifies the degree to which the algorithm's apparent performance on imbalanced data is driven by majority-class dominance rather than genuine discriminative capability. Dataset-2 results are therefore the appropriate basis for comparing Logistic Regression's utility against other classifiers.
- Naive Bayes is the weakest classifier on both datasets (78.92% and 63.21%), consistent with the theoretical expectation that conditional feature independence — the algorithm's core assumption — is severely violated by the strong co-linear relationships among gas supply, LNG import volume, and load-shedding in the present dataset.
- The Apriori algorithm yielded actionable operational rules: Rule R1 established with 94.1% confidence that the co-occurrence of a gas shortage and Summer season is almost certain to produce Severe load-shedding. This rule is immediately deployable as a two-variable early warning heuristic by BPDB grid operators, requiring no machine learning infrastructure.
- k-Means clustering identified four energy regime clusters whose centroids align closely with the three target severity classes, providing data-structural validation that the class boundaries defined in Section 3.2 are empirically well-grounded and that the load-shedding severity classification problem is well-posed.

From a policy perspective, the results indicate that a Decision Tree classifier trained on publicly available BPDB and BPC data could serve as an operational early warning tool for Severe load-shedding events in Bangladesh, enabling grid managers to pre-position fuel reserves and issue public notices one to seven days in advance. The Apriori rules offer a complementary, interpretable heuristic that does not require a deployed model and can be communicated directly to non-technical stakeholders.

Several directions for future research are identified. First, ensemble methods — particularly Random Forest, Gradient Boosting, and XGBoost — should be evaluated to assess whether combining multiple weak classifiers can push classification accuracy beyond the 93.81%

Decision Tree baseline, especially for the minority Severe class. Second, the classification framework could be extended to a 24-hour-ahead probabilistic forecast formulation using time-lagged features (e.g., lagged load-shedding values, ARIMA residuals of fuel stock trends) to make the model suitable for actual operational deployment. Third, division-specific models — rather than a single pooled model — should be explored, given the markedly heterogeneous load-shedding profiles observed across divisions (Table 5). Fourth, incorporating satellite-derived weather data and near-real-time BPC fuel import tracking data could substantially enrich the feature set. Finally, the deployment of the best-performing model as a web-accessible decision-support dashboard for BPDB operations staff represents a natural and high-impact applied extension of this work.

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Appendix A: Variable Description Tables

Table A.1: Explanatory Variables Utilised by Each Algorithm

Variable	DT	NB	LR	NN	KNN	Apriori	k-Means	Target
Peak_Demand_MW	✓	✓	✓	✓	✓		✓	
Peak_Generation_MW	✓	✓	✓	✓	✓			
Temperature_Dhaka	✓	✓	✓	✓	✓		✓	
Gas_Shortage_Flag	✓	✓	✓	✓	✓	✓		
Coal_Shortage_Flag	✓	✓	✓	✓	✓	✓		
Low_Water_Flag	✓	✓	✓	✓	✓			
Season (encoded)	✓	✓	✓	✓	✓	✓		
Division (encoded)	✓	✓	✓	✓	✓	✓		
Day_of_Year	✓	✓	✓	✓	✓			
Furnace_Oil_Stock_MT	✓	✓	✓	✓	✓			
Diesel_Stock_MT	✓	✓	✓	✓	✓			
LNG_Import_MMCFD	✓	✓	✓	✓	✓		✓	
Domestic_Gas_MMCFD	✓	✓	✓	✓	✓			
Generation_Capacity_MW	✓	✓	✓	✓	✓			
Target_Severity						✓		✓

Table A.2: Administrative Divisions of Bangladesh — Key Statistics

Division	Area (km²)	Population (M)	Dominant Economic Activity	Typical Peak Demand (MW)
Dhaka	20,594	36.0	Garment manufacturing, financial services, commerce	4,800 – 5,500
Chattogram	33,909	28.4	Seaport operations, heavy industry, export processing	2,200 – 2,800
Rajshahi	18,174	19.5	Agriculture, silk production, ceramic manufacturing	1,400 – 1,900
Khulna	22,285	15.6	Shrimp aquaculture, jute processing, industrial zones	1,100 – 1,600

Barishal	13,297	8.3	Agriculture, inland fisheries, river transport	600 – 900
Sylhet	12,596	10.0	Tea cultivation, overseas remittances, natural gas fields	700 – 1,100
Rangpur	16,317	15.7	Agricultural production, tobacco processing	1,000 – 1,400
Mymensingh	10,485	11.4	Agriculture, small-scale manufacturing, agro-processing	900 – 1,200

Appendix B: Description of Submitted Files

Table B.1: Files Submitted with this Project

File Name	Description and Purpose
Project_Report.pdf	This project report document summarising the methodology, results, and findings.
load_shed_classification.ipynb	Implements all five supervised classifiers: Decision Tree, Naive Bayes, Logistic Regression, Neural Network (MLP), and K-Nearest Neighbour. Includes 10-fold cross-validation and confusion matrix generation.
load_shed_apriori.ipynb	Implements the Apriori association rule mining algorithm using the mlxtend library on the binarised transaction dataset. Outputs support and confidence tables for all rules meeting threshold criteria.
load_shed_clustering.ipynb	Implements k-Means clustering with Elbow Method selection of k, cluster centroid reporting, and 3-D scatter plot visualisation of cluster assignments.
undersampling.ipynb	Produces Dataset-2 from Dataset-1 by random undersampling of the None and Moderate majority classes to achieve balanced class proportions.
dataset1_bpdb_clean.csv	Dataset-1: 14,936 records, 18 input features, post-cleaning. Primary input for all classification and clustering experiments.
dataset2_balanced.csv	Dataset-2: 6,482 records, balanced across the three severity classes. Used for all supervised classifier experiments alongside Dataset-1.
dataset_apriori.csv	Binarised transaction-format version of Dataset-1 prepared for use as input to the Apriori algorithm.
EDA_analysis.xlsx	Exploratory data analysis workbook containing pivot tables, time-series charts, class-conditional feature distributions, and correlation matrices.
bpdb_raw_scraped.csv	Original raw dataset scraped from the BPDB digital archive (Version 1, unprocessed). Provided for full reproducibility.
bpc_fuel_stocks_2019_2024.csv	Monthly BPC fuel stock data and Petrobangla natural gas supply figures for 2019–2024, used to construct the supplementary fuel-side features in the merged dataset.